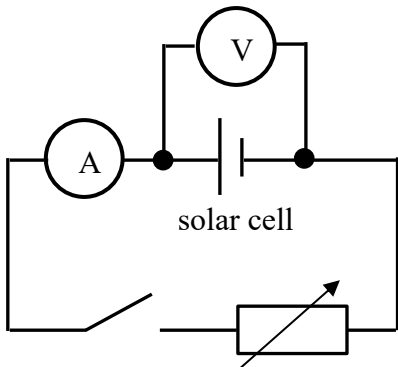


4	
(a)(i)1.	effective resistance = $(\frac{1}{600} + \frac{1}{3000})^{-1} + 30$ = 530 Ω
	terminal p.d. = $\varepsilon - Ir$ $= 12.0 - (\frac{12.0}{530})(30)$ $= 11.3 \text{ V}$ $I_{\text{LDR}} = \frac{11.3}{3000}$ $= 3.37 \text{ mA}$
	= 3.8 mA (2sf) (shown)
(a)(i)2.	$P_{\text{LDR}} = I_{\text{LDR}}^2 R$ $= (3.8 \times 10^{-3})^2 (3000)$
	= 0.043 W (2sf)

Qn	Suggested Answer
(a)(ii)	new effective resistance = $\left(\frac{1}{600} + \frac{1}{100}\right)^{-1} + 30$ = 116 Ω new terminal p.d. = $\varepsilon - Ir$ = $12.0 - \left(\frac{12.0}{116}\right)(30)$ = 8.90 V
	new $P_{LDR} = \frac{V_{LDR}^2}{R}$ = $\frac{8.90^2}{100}$ = 0.79 W (2sf) Since $P_{LDR} > 0.50$ W, it will be damaged.
(a)(iii)	Light bulb drawn - in series with 600 Ω resistor OR - parallel to LDR and 600 Ω resistor.
	When light intensity is low, resistance of LDR is high. <u>Total resistance</u> in circuit will <u>increase</u>, leading to a smaller total current. This would lead to a smaller drop in p.d. across the internal resistance and a <u>higher p.d. across light bulb</u>.
(b)(i)	 Solar cell, voltmeter and ammeter drawn and connected correctly. Variable resistor to vary resistance in circuit.
(b)(ii)1.	When output current is zero, output p.d. = $E = 0.580$ V
(b)(ii)2.	$R = \frac{V}{I}$ = $\frac{0.550}{20.0 \times 10^{-3}}$ = 27.5 Ω (3sf)

Qn	Suggested Answer
(b)(ii)3.	$V = E - Ir$ $r = \frac{E - V}{I}$ $= \frac{0.580 - 0.550}{20.0 \times 10^{-3}}$
	= 1.5 Ω

Qn	Suggested Answer
5	